

BEC503 Digital Communication — Assignment 4 —

Linear block and cyclic code construction

Module 4: Error Control Coding: Linear Block and Cyclic Codes

Field	Value
Maximum marks	20
Weightage	10% of CIE
Issue date	<<TBD>>
Due date	<<TBD>>
Items	8 problems + 3 theory questions
CO / Bloom coverage	CO5: 7, CO4: 4 · L1–L2 0, L3–L4 5, L5–L6 6

Objective

Build codes rather than describe them: construct generator and parity check matrices, decode with syndromes, and design a shift-register encoder that a technician could wire up.

Learning outcomes

- Student will be able to construct G and H for a given linear block code and verify their relation.
- Student will be able to compute minimum distance and state the exact detection and correction guarantees.
- Student will be able to build a standard array and decode by syndrome lookup.
- Student will be able to design and trace a shift-register encoder for a cyclic code.

Part A — Problems (17 marks)

P1. A (6,3) linear block code has $P = [[1,1,0],[0,1,1],[1,0,1]]$. Write G and H in systematic form, list all eight codewords, and compute the minimum distance by weight enumeration. State the detection and correction capabilities. [C05 | L4 | 3 marks]

P2. For the code of P1, construct the complete standard array. Identify every coset leader, tabulate the syndrome for each, and state how many of the fifteen possible double-error patterns the code corrects. [C05 | L4 | 3 marks]

P3. Construct the parity check matrix of a (15,11) Hamming code with columns ordered as binary counting. Encode any message of your choice, introduce a single error, and demonstrate that the syndrome read as a binary number gives the error position. [C05 | L4 | 3 marks]

P4. Factor $x^{15} - 1$ over $GF(2)$. List every cyclic code of length 15 that can be built from its factors, giving $g(x)$, k , the rate and, where you can determine it, $d_{\min}$. Identify which is the Hamming code. [C05 | L5 | 3 marks]

P5. Design the shift-register encoder for the (7,4) cyclic code with $g(x) = x^3 + x + 1$. Draw the circuit, then trace the register contents clock by clock for the message 1101 and confirm the parity bits match a

hand division. [C05 | L5 | 2 marks]

P6. For that same code, tabulate the syndrome for every single-error pattern. Then receive the word 0111001, compute its syndrome, correct it, and extract the message. [C05 | L4 | 1 marks]

P7. Show that the (7,4) cyclic code detects every burst of length 3 or less. Then find a burst of length 4 that it fails to detect and explain why that particular pattern escapes. [C04 | L5 | 1 marks]

P8. A system needs to correct two errors per 15-bit block. Explain why no Hamming code can do this and identify what family of codes would be required. [C04 | L5 | 1 marks]

Part B — Theory (3 marks)

T1. Argue whether systematic form is ever a disadvantage, considering both error performance and any application where an exposed message matters. [C05 | L4 | 1 marks]

T2. Cyclic codes trade generality for structure. Identify precisely what is given up by requiring $g(x)$ to divide $x^n - 1$, and what is gained in return. [C04 | L5 | 1 marks]

T3. A zero syndrome does not prove the absence of errors. Characterise exactly which error patterns escape detection, and estimate how often that happens on a channel with bit error rate $1e-3$. [C04 | L5 | 1 marks]

Submission guidelines

- Format: PDF, single file
- File name: <USN>_BEC503_Assignment4.pdf
- Channel: LMS upload
- Late policy: 10% per day, nothing after 5 days
- Every numerical answer must carry units and a stated reference direction.

Assessment rubric

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Correctness of matrix and polynomial construction	8	All matrices and polynomials correct and verified against their defining relations.	Correct with minor unverified steps.	Constructions attempted with several errors.	Constructions largely wrong or absent.
GF(2) arithmetic accuracy	5	All modulo-2 arithmetic correct throughout, including long division.	One or two slips not affecting conclusions.	Frequent arithmetic errors.	Arithmetic unreliable; ordinary integer arithmetic used in places.
Circuit design and traces	4	Encoder drawn correctly with taps in the right places and the trace verified against hand division.	Circuit correct, trace incomplete.	Circuit drawn with tap errors.	No circuit or an unworkable one.

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Reasoning about guarantees	3	States exactly what each code guarantees and where the guarantee ends.	States guarantees correctly without discussing limits.	Guarantees quoted imprecisely.	Guarantees misstated.
Total	20				

Course outcome attainment

Course outcome	Marks carried
CO4	9
CO5	11
Total	20

Academic integrity

Individual submission. Identical submissions score zero for both parties.